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Internal Camera-based Reconstruction and Closed-loop Control of Soft Robotic Arms

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ABSTRACT

Soft continuum robotic arms offer unparalleled dexterity and compliance. However, their virtually infinite degrees of freedom and nonlinear deformations pose significant challenges for sensing and control. This paper proposes a vision-based framework for distributed state estimation and closed-loop control of tendon-driven soft robotic arms. Miniature cameras embedded in each section provide dense proprioceptive feedback, while convolutional neural networks map multi-view images to both tip pose and full-body reconstruction using a polynomial curvature model. Two alternative controllers are designed and evaluated. The first is a model-based null-space velocity controller that regulates tip pose while shaping tendon configurations. The second is a data-driven inverse kinematics neural network that maps task-space goals to tendon commands, with explicit secondary-objective optimization during training. Both approaches are validated in open- and closed-loop settings on single- and multi-section manipulators in quasi-static regimes. Experiments demonstrate that embedded vision enables accurate full-body state reconstruction, reliable redundancy resolution, and robust regulation even under external disturbances.

1 | Introduction

Soft robots draw inspiration from natural organisms to achieve compliance and distributed motion [1]. They are designed to passively deform and reconfigure to navigate complex environments and adapt to their surroundings. This embodied intelligence facilitates compliant coupling with external objects, stemming from variable impedance characteristics [2] and potentially infinite degrees of freedom. Among soft robots, continuum manipulators are characterized by their slender configuration, and capacity to have complex 3D curvatures. Due to their adaptability and safe interaction, they show promise in applications such as surgical tools for minimally invasive diagnosis [3], dexterous grasping and manipulation [4], and environmental monitoring and exploration [5]. However, they still lack precision and robustness to external disturbances [6], requiring dense sensing and redundancy-aware control.

Accurate proprioceptive control in soft arms depends on adequate feedback, as they are strongly affected by deformations induced by environmental interactions, self-weight, and approximate modeling [7, 8]. Sensing technologies face major challenges, both in embedding sensors throughout the body and in reconstructing the end-effector pose from indirect measurements. One approach is to directly detect skin deformations using resistive or piezoresistive strain and pressure sensors, or Fiber Bragg Grating (FBG) nodes [5]. Another solution is to integrate Inertial Measurement Units (IMUs) into the arm, enabling high-frequency pose reconstruction while avoiding the occlusion and calibration issues of external sensing [6]. IMUs also allow estimation of body deformation under external forces [9].

However, IMU sensing suffers from drift [10] and, being non-distributed, cannot capture the global strain state. Camera-based approaches are increasingly used in soft robotics for pose reconstruction, as they can capture both global deformations and local interactions [11], while avoiding the integration challenges of embedded strain or pressure sensors [5]. External setups have been proposed, including stereometric systems for supervised shape learning [12] and model-free approaches for closed-loop full-body control of soft robots [7]. These methods nevertheless become

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impractical in the presence of occlusions, and the cameras must remain fixed.

For proprioceptive performance, the sensing system should track the robot motion from internal cameras, without interfering with movement or compliance of the structure. Previous works have embedded cameras inside the hollow body, often employing internal illumination such as structured light [13] to ensure consistent imaging. In this framework, a common approach is to place visual markers beneath the robot skin. To further improve robustness, image binarization is often applied before inference [14], and simulation is used to pre-train models for zero-shot transfer to the real world [15]. Approaches for reconstruction include geometric optimization from images [16] and deep learning architectures for state mapping. Embedded cameras provide high-resolution and accurate measurements of global deformations, without requiring complex distributed sensing hardware. Due to these advantages, vision-based strategies have been exploited for contact and displacement detection [11], tactile servoing [17], and force estimation [18].

Onboard state estimation of soft structures allows for the deployment of various control architectures. Yet, controlling continuum arms presents significant challenges. These systems are inherently redundant [1] and therefore offer multiple possibilities to configure their actuation space. They possess extra degrees of freedom beyond those required to accomplish the primary task, a property referred to as kinematic redundancy in the null space. This redundancy enables the optimization of secondary objectives, such as minimizing control effort, improving manipulability, and avoiding obstacles or singularities [19]. Prior works have applied null-space control of soft robots, employing path planning algorithms such as RRT [20] to randomize the robot configuration while reaching a goal, or teleoperating the tip of a continuum manipulator with a rigid-link master to maximize manipulability [21]. More advanced objectives, such as whole-body grasping [22], have been demonstrated by assigning and regulating task-space coordinates at several locations along the backbone. However, validation of these approaches has been limited to simulation. Alternatively, researchers have controlled tendon-driven origami soft robots by implicitly solving redundant inverse kinematics through constrained gradient-based optimization, using motion capture for feedback [23].

Control strategies for continuum manipulators can be divided into model-based and data-driven approaches [1]. Many model-based controllers adopt a spatial backbone representation that neglects volumetric deformations and focuses exclusively on the behavior of the central axis [24]. A general kinematic formulation is the Piecewise Constant Curvature (PCC) model [25], which approximates the robot as a sequence of circular arcs. Variants of this model have also been developed for Jacobian-based control [6]. More expressive alternatives include Cosserat rod models, which capture shear and torsional deformations through a geometrically exact formulation [26]. FEM-based frameworks further leverage continuum mechanics for simulation and control, at the cost of higher computational demand [27]. Data-driven controllers address the limitations of these approaches, including non-idealities in soft robotics such as friction, material anisotropy, and cable slack [1]. One direction is adaptive control, where an online learning loop updates model parameters from sensor feedback [28]. Another is iterative learning control, where the robot improves its behavior by incorporating errors from previous executions into the feedforward command while preserving its natural compliance [2]. Alternatively, machine learning approaches can approximate the entire robot model. Researchers have developed a biLSTM-based framework for planning and control that relies only on motor encoder feedback [10].

Addressing the sensing and control challenges of continuum manipulators, this paper proposes a strategy for vision-based reconstruction and control of soft robotic arms that scales with the number of segments. Information from miniature cameras embedded in the sections is fused within a single CNN to achieve full-body reconstruction. The reconstructed state is then used as feedback in both model-based and data-driven controllers, enabling tip regulation while reducing actuation effort. To the best of our knowledge, no framework has yet combined such sensing with closed-loop null-space control of continuum robots, either through model-based or learning-based approaches.

The contributions of this paper include:

1. A distributed and scalable vision-based sensing strategy for soft arms that uses miniature cameras embedded in each section of the robot. It provides dense, markerless feedback for full-body state reconstruction, and scales to arms with different numbers of sections through modular hardware design and a lightweight CNN architecture.
2. Integration of embedded vision with model-based null-space control. We close the loop by combining camera-based state estimation with a Jacobian-based Piecewise Constant Curvature velocity controller.
3. A data-driven framework for both sensing and control. We train learning-based inverse kinematics models to generate actuation commands while optimizing redundancy objectives.
4. Experimental validation of precise pose regulation using both model-based and data-driven controllers, and demonstration of the advantages of closed-loop control under external disturbances.

In the remainder of this paper, Section 2 describes the robot setup, the curvature models, and the data acquisition strategy. Section 3 presents the deep learning methods and the control architectures. Section 4 outlines the experimental results. Section 5 summarizes the paper and discusses potential directions for future work. A formal definition of all notation and terminology used throughout these sections is provided in the Supporting Information.

2 | Setup and Specification

2.1 | Experimental Setup

In this work, we used two compressible soft robotic arms setups for the experiments: a multi-section arm configuration, shown in Fig. 1, and a single-section arm configuration which corresponds to the first segment of the multi-section arm. The multi-section robot comprises three independent modules, each 150 mm long and with a 60 mm diameter, based on the Trimmed Helicoid (TH) structure [29]. Each arm module is actuated by three motors (DYNAMIXEL xl330-m288-t) mounted on the robot base and connected via three cables arranged at 120° around the circumferential direction. The top of each section is rigidly attached to the base of the next, and motor rotations allow both bending and

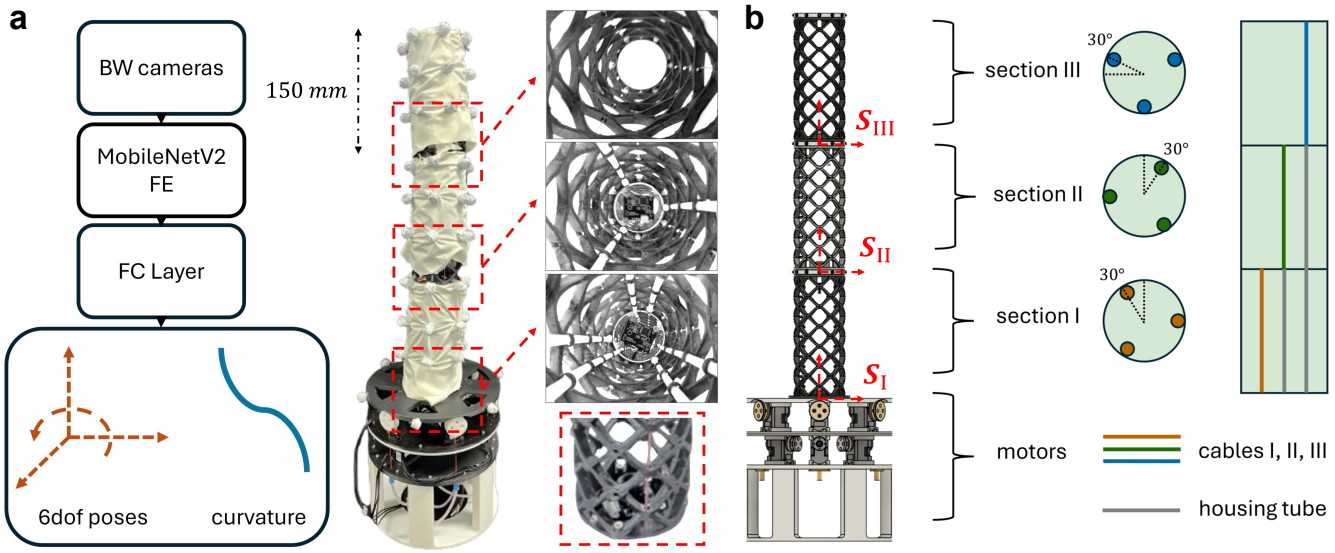

FIGURE 1 | (a) Multi-section soft robot arm sensorization and reconstruction pipeline. Images from the three cameras are acquired simultaneously and fed to CNNs, which predict both the tip poses and the full-body curvature. (b) Robot setup. The tendon-driven arm consists of three independent segments, each actuated by three motors, with Bowden cables routing the tendons of sections II and III. The optical tracking system provides the ground-truth configuration.

compression through 6 mm-diameter pulley wheels. Three motors actuate the first section directly, while six drive the remaining two via Bowden cables such that their actuation is decoupled. The cables of Module II and III are rotated with respect to Module I of 60° and 150°, respectively. The base, pulleys, and camera supports were 3D printed in PolyLactic Acid (PLA), while helicoidal components in Thermoplastic PolyUrethane (TPU).

Internal cameras are added to each section for proprioception, and are rigidly mounted between sections and at the robot base. Consequently, each view is fixed relative to the base of the subsequent section. The sensing setup has negligible impact on the compliance of the soft structure. Camera images are transmitted to a laptop via three USB cables at a resolution of 480 × 640 in grayscale. No internal markers, fiducial tags, or structured light were used in this setup, but proprioception exploits the changing internal geometry of the structure with deformation. Camera views from the three sections are shown in Fig. 1**a**. To avoid interference from external light, the robot arm sections are covered with a diffusive layer attached to the skin. We tested two different solutions: the single-section robot is covered with corrugated silicone to accommodate compression, whereas the multi-section arm employs a stretchable white fabric. Since both covering materials have negligible stiffness compared to the robot structure, we assume equivalent kinematics of each section.

To capture ground-truth data for the robot's position and curvature, optical tracking markers are fixed at the base, at the end of each module, and at $\frac{1}{3}$ and $\frac{2}{3}$ of each section. A laptop receives tracking data from the Motive computer via an Ethernet switch and communicates with the motors through a U2D2 power hub board (DYNAMIXEL). Before each data acquisition session or experimental trial, the robots are calibrated by tightening all tendons through current control of the actuation motors. The motors then operate in position-control mode with a predefined current limit.

2.2 | Problem Formulation

To enable model-based control and full-body proprioception, we formulate two kinematic models: a simplified one for control and a more expressive one for shape reconstruction.

The model-based controller relies on the Piecewise Constant Curvature (PCC) kinematic approximation of soft continuum robots. Under the PCC assumption, each segment bends in any direction while maintaining constant curvature, resulting in a continuously differentiable backbone curve. Each actuation section is parameterized as shown in Fig. 2.

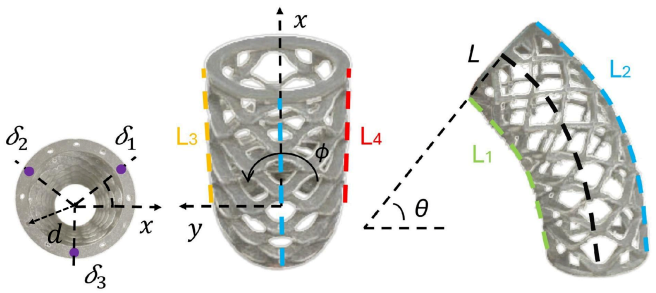

FIGURE 2 | PCC parameterization of soft robots according to both $\alpha\,(\theta, \phi, L)$ and $q\,(\Delta_x, \Delta_y, L)$ parameters, derived from the four arc lengths $\{L_i\}_{i=1,\dots,4}$. The tendon angles $\{\delta_i\}_{i=1,\dots,3}$ and the radius d are also indicated.

The α-parameterization for this robot is defined by three parameters: the bending angle θ, the bending direction ϕ, and the central axis length L. However, this transformation is not an isomorphism and may lead to Jacobian rank loss [30]. We therefore adopt the q-parameterization $q = [\Delta_x, \Delta_y, L]^{\mathrm{T}}$, where $\Delta_x = \theta d \cos\phi$ and $\Delta_y = \theta d \sin\phi$ are base arc-length differences derived from the four arc lengths L_1, L_2, L_3, L_4 (Fig. 2), and d is the

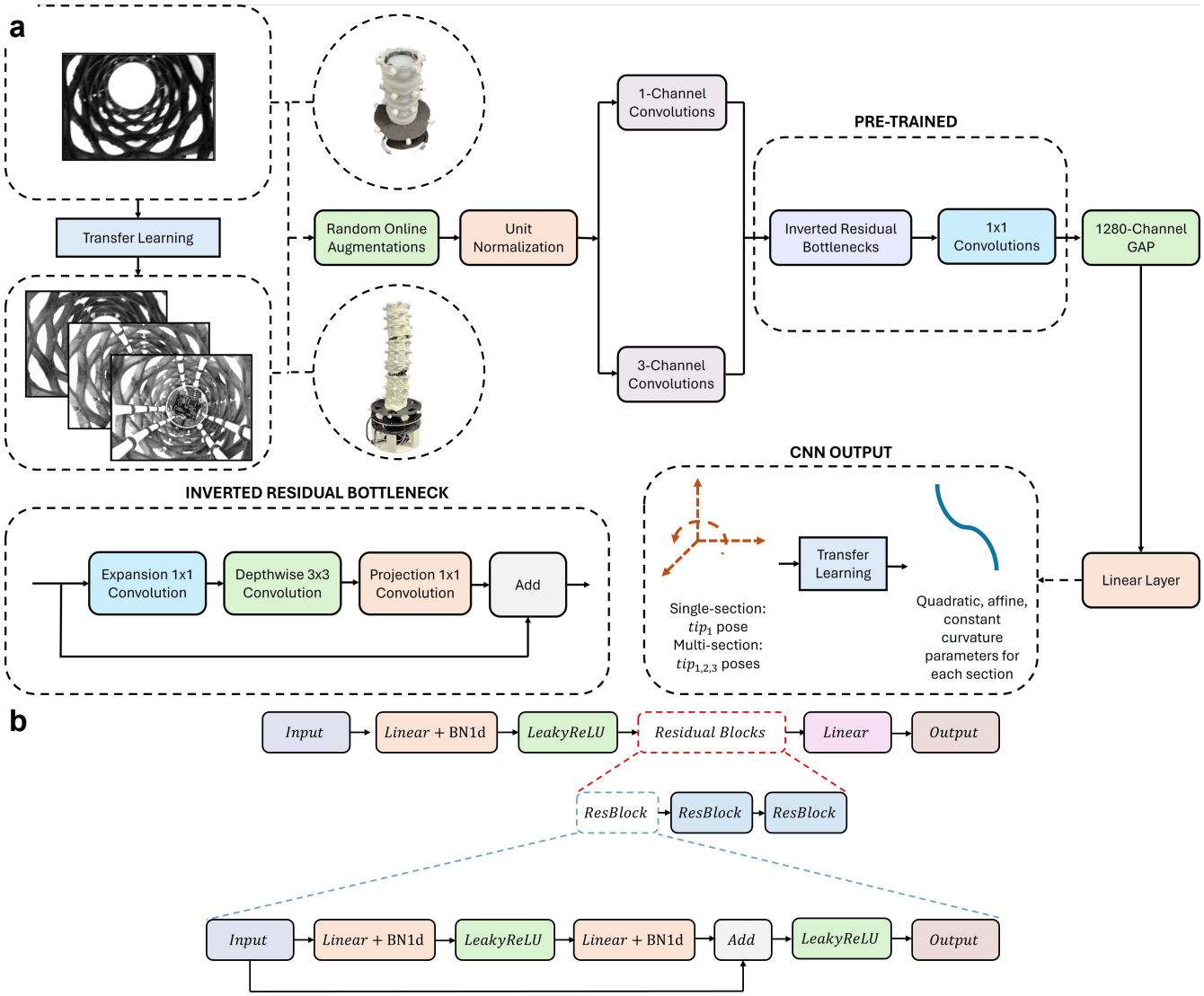

FIGURE 3 | (a) Convolutional neural network (CNN) architecture used for full-body reconstruction. Stacked images from embedded cameras are processed through data augmentation and custom convolutional kernels. A MobileNetV2 backbone with inverted residual bottlenecks is fine-tuned on our dataset and followed by global average pooling (GAP) to produce a compact feature vector. This representation is used to predict tip poses (6 outputs for the single-section, 18 for the multi-section) or curvature parameters for each section, using dedicated networks for different approximations. Transfer learning is applied between pose and curvature networks and to adapt the single-section to the multi-section network. (b) Multi-layer perceptron (MLP) architecture used for kinematics approximations. The network consists of an input projection layer, residual blocks, and an output regression head. The input vector is projected into a 128-dimensional latent space and processed through three residual blocks, each comprising two fully connected layers with Batch Normalization and LeakyReLU activations. Skip connections enable residual learning and help mitigate vanishing gradients.

section radius. Defining the tendon lengths as $l = [l_1, l_2, l_3]^{\mathrm{T}}$, with tendons at angles δ_i:

$$l_i = L - \theta d \cos(\delta_i - \phi) = L - \Delta_x \cos\delta_i - \Delta_y \sin\delta_i \tag{1}$$

yielding the mapping between l and q as:

$$\Delta_x = -\frac{2}{3}\sum_{i=1}^{3} l_i \cos\delta_i,\ \Delta_y = -\frac{2}{3}\sum_{i=1}^{3} l_i \sin\delta_i,\ L = \frac{1}{3}\sum_{i=1}^{3} l_i \tag{2}$$

Let $\Delta = \sqrt{\Delta_x^2 + \Delta_y^2}$ and $T \in SE(3)$ the homogeneous transformation mapping the base frame to the tip. The rotation matrix and translation are (with $c_{\frac{\Delta}{d}} = \cos\frac{\Delta}{d}$, $s_{\frac{\Delta}{d}} = \sin\frac{\Delta}{d}$):

$$R = \begin{bmatrix} 1 + \frac{\Delta_x^2}{\Delta^2}\left(c_{\frac{\Delta}{d}} - 1\right) & \frac{\Delta_x\Delta_y}{\Delta^2}\left(c_{\frac{\Delta}{d}} - 1\right) & \frac{\Delta_x}{\Delta} s_{\frac{\Delta}{d}} \\ \frac{\Delta_x\Delta_y}{\Delta^2}\left(c_{\frac{\Delta}{d}} - 1\right) & 1 + \frac{\Delta_y^2}{\Delta^2}\left(c_{\frac{\Delta}{d}} - 1\right) & \frac{\Delta_y}{\Delta} s_{\frac{\Delta}{d}} \\ -\frac{\Delta_x}{\Delta} s_{\frac{\Delta}{d}} & -\frac{\Delta_y}{\Delta} s_{\frac{\Delta}{d}} & c_{\frac{\Delta}{d}} \end{bmatrix} \tag{3a}$$

$$t = \frac{dL}{\Delta^2}\left[\Delta_x\left(1 - c_{\frac{\Delta}{d}}\right)\ \Delta_y\left(1 - c_{\frac{\Delta}{d}}\right)\ \Delta s_{\frac{\Delta}{d}}\right]^{\mathrm{T}} \tag{3b}$$

We choose to express section rotations as Euler angles $(\psi_{\text{yaw}}, \theta_{\text{pitch}}, \phi_{\text{roll}})$ for their interpretability; gimbal lock cannot occur since torsional motion is impossible in these sections.

However, soft arms undergo deformations due to external forces and self-weight, requiring more complex shape approximations for full-body proprioception. We adopt the Piecewise Polynomial Curvature (PPC) model [31], which retains the α-parameters. It expresses the curvature $c(s)$ of the central axis of a segment as a polynomial in the normalized arc length $s \in [0, 1]$, with the local bending angle $\theta(s) = \int_0^s c(v)dv$. For example, in the quadratic case:

$$\theta(s) = c_0 s + \frac{1}{2}c_1 s^2 + \frac{1}{3}c_2 s^3 \tag{4}$$

Let $T^s \in SE(3)$ denote the homogeneous transformation from the base frame to any point along the section axis (with $c_\phi = \cos\phi$, $s_\phi = \sin\phi$, $c_{\theta(s)} = \cos\theta(s)$, $s_{\theta(s)} = \sin\theta(s)$):

$$R^s = \begin{bmatrix} c_\phi^2(c_{\theta(s)} - 1) + 1 & s_\phi c_\phi(c_{\theta(s)} - 1) & c_\phi s_{\theta(s)} \\ s_\phi c_\phi(c_{\theta(s)} - 1) & s_\phi^2(c_{\theta(s)} - 1) + 1 & s_\phi s_{\theta(s)} \\ -c_\phi s_{\theta(s)} & -s_\phi s_{\theta(s)} & c_{\theta(s)} \end{bmatrix} \tag{5a}$$

$$t^s = \begin{bmatrix} c_\phi \int_0^s L s_{\theta(v)} dv & s_\phi \int_0^s L s_{\theta(v)} dv & \int_0^s L c_{\theta(v)} dv \end{bmatrix}^{\mathrm{T}} \tag{5b}$$

For capturing ground truth pose, reflective markers for motion capture are placed along each segment at intervals of $\frac{1}{3}$ of its length. We refer to these bodies as *subsections* of the module. They provide ground-truth bending direction and angle estimates $\hat{\phi}_s$, $\hat{\theta}_s$ at $s = \frac{1}{3}, \frac{2}{3}, 1$ (details in the Supporting Information). We adopt the optimization in [9], applied identically to quadratic, affine, and constant models. Coefficients are initialized analytically from these measurements solving proper linear systems. Starting from these guesses, we optimize the coefficients by minimizing the deviation between predicted and measured rotation matrices. Denoting by $\|\cdot\|_F$ the Frobenius norm, the quadratic curvature parameters are:

$$\arg \min_{c_0, c_1, c_2 \in \mathbb{R}} \sum_{s=\frac{1}{3}, \frac{2}{3}, 1} \left\| R^s(c_0, c_1, c_2) - \hat{R}^s \right\|_F^2 \tag{6}$$

The bending direction ϕ is the mean of $\hat{\phi}_{\frac{1}{3}}, \hat{\phi}_{\frac{2}{3}}, \hat{\phi}_1$, while L is the average tendon length according to Eq. 2.

3 | Vision Based Reconstruction and Control

3.1 | Vision Based Proprioception

To reconstruct the kinematics of the robot from the internal cameras, we use two deep learning architectures. The first is a Convolutional Neural Network (CNN) to estimate pose and curvature parameters directly from camera images. The second is a Multi-Layer Perceptron (MLP) for modeling the forward and inverse kinematics of both robotic arms.

We use MobileNetV2 [32] as a pre-trained CNN backbone. This lightweight architecture reduces training effort, mitigates overfitting, and limits the sensing frequency to about 8 Hz, which is sufficient for quasi-static control. The input consists of stacked grayscale images from the cameras placed along the manipulator, with a shape of $480 \times 640 \times n$, where n denotes the number of sections of the soft arm. To process the image batches, the first 2D kernel is modified and preceded by a normalization layer. This is the only non-pre-trained component of the feature extractor, making the sensing design scalable, as adapting to a different number of sections requires adjusting only the first layer (Fig. 3**a**). At the end of the backbone, we apply Global Average Pooling (GAP) followed by a fully connected head that maps directly to the regression outputs. Let I denote the stacked images from the BW cameras. We denote by f_p^1 and f_C^1 the networks mapping I to the tip pose and full-body curvature parameters of the single-section arm, and f_p^3 and f_C^3 for the multi-section arm.

Forward and inverse kinematics are modeled using a custom MLP architecture, detailed in Fig. 3**b**. The network utilizes residual blocks to prevent vanishing gradients, yielding the inverse mappings $N_l^1 : t \mapsto l$ and $N_l^3 : p \mapsto l$, and the forward mapping $N_p^3 : l \mapsto p$.

3.1.1 | Network Training

All networks were trained on an NVIDIA GeForce RTX 3090 GPU. We used transfer learning within the same robot and across the two manipulators: f_C^1 networks were initialized with the f_p^1 backbone, and f_p^3 was initialized from f_p^1. To improve generalization, we applied online data augmentation simulating changes in illumination and brightness to account for residual variations through the diffusive skin. The dataset was split into training (85%), validation (10%), and test (5%) sets. Training employed the Adam optimizer with $\eta = 10^{-3}$, weight decay, and early stopping. We employed Huber loss ($\mathcal{L}$) to train all regressors to compensate for ground-truth noise and outliers [33].

3.1.2 | Model Spatial Attention

Inspired by [15], we adapt the Grad-CAM method [34] to visualize the image regions that most influence the prediction (details in the Supporting Information). As illustrated in Fig. 4**a**, the network generalizes by focusing primarily on structural deformations of the trimmed helicoid rather than the tip itself, which is not always fully visible. Specifically, in bent configurations, the network heavily relies on the elongated side opposite the bending direction while also exploiting the compressed side. In straight configurations, attention shifts toward the four corners where structural variations are strongest, enabling robust state estimation across both bending and compression. The average activation map confirms this pattern, with consistent attention concentrated at the four corners across the dataset.

3.2 | Workspace Sampling

The CNN and MLP architectures require supervised training data spanning the workspaces of the manipulators. We collected 10,000 data points for both single- and multi- section configurations using six Optitrack Prime 13 cameras and Motive software, covering bending angles up to 85° per section.

All pose information is transformed to the base origin reference frame (frames detailed in Fig. 1**b**), yielding a generic pose vector $p = [t, r]^{\mathrm{T}}$ (position $t = [x, y, z]^{\mathrm{T}}$, orientation $r = [\psi_{\text{yaw}}, \theta_{\text{pitch}}, \phi_{\text{roll}}]^{\mathrm{T}}$). Curvature parameters C_j for segment j are defined in the local module base frame (e.g., $C_j = [c_0, c_1, c_2, L, \phi]_j^{\mathrm{T}}$ for quadratic models). Tendon lengths for each section are denoted $l_j = [l_{1j}, l_{2j}, l_{3j}]^{\mathrm{T}}$; for the multi-segment case, $j \in \{I, II, III\}$, forming the full actuation vector $l = [l_I, l_{II}, l_{III}]^{\mathrm{T}}$. For each data point, we captured camera images, module poses (tip

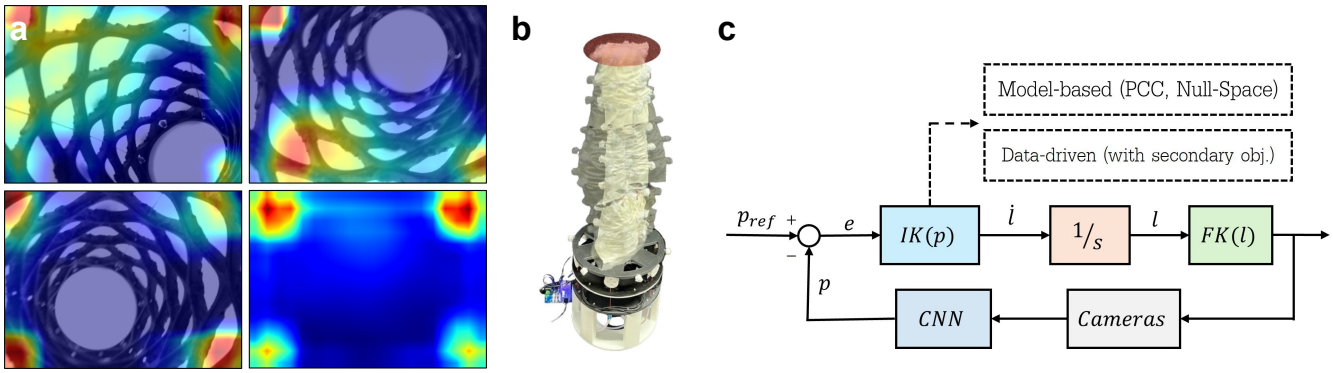

FIGURE 4 | (a) Grad-CAM visualizations from f_p^1 for different bending configurations and average Grad-CAM over the test set. The network mostly focuses on deformations of the Trimmed Helicoid structure along the bending direction. (b) Different null-space configurations yield the same tip pose. (c) High-level kinematic closed-loop controller employing CNN-based feedback. Tendons can be actuated using either model-based or data-driven approaches.

pose p, and intermediate poses p_1, p_2 for the multi-segment case), tendon lengths l, and segment curvature parameters $C = \{C_j\}$ across quadratic, affine, and constant curvature models.

While the single-section robot sampling utilized pure motor babbling, the multi-section arm requires a structured strategy to explore workspace boundaries where modules bend congruently. We adapted the strategy in [10] for a compressible robot by defining a *Bending Solver BS* : $l_j \mapsto l_k$ to map tendon lengths between sections to achieve identical q-configuration. The sampling procedure is statistically balanced: one-third of the samples actuate all modules to the same configuration, one-third couple two modules while the remaining one is independent, and the final third utilizes full independent actuation.

3.3 | Controller Design

For the single-section robot, a controller is implemented to reach a desired position, while the pose is controlled for multi-section case. The multi-section arm is an overactuated system with nine distributed actuators and six control degrees of freedom. An example of the corresponding 3-DOF null space is shown in Fig. 4**b** demonstrating how different structural configurations, whether compressed or bent, yield the same tip position and orientation. The manipulator can be shaped within its kinematics to exploit these equivalent states.

All the designed kinematic controllers follow the same general scheme shown in Fig. 4**c**. A trajectory planner $\mathcal{W}^k$ (where k is the number of sections) generates a time-varying reference pose or position. The error between the reference and the current state, estimated in closed-loop by the CNN from real-time camera input, is then fed to an inverse kinematics (IK) solver. This updates the tendon actuations over time, driving the robot through its actual forward kinematics (FK), until the distance to the goal falls below a given threshold γ.

3.3.1 | Single-Section Controller

The single-section controllers operate on a fully actuated system. At each time step, the system is locally linearized through its Jacobian to compute the tendon actuation required to reach t_{ref}. For the model-based controller, vision-based sensing is combined with encoder position feedback. Once l is measured from the motor encoders, the forward derivative J is computed as $J_{t2l}(l)$. To mitigate poor conditioning, we employ its regularized inverse $J^{\#} = J^T(JJ^T + \lambda^2 I)^{-1}$, and compute the tendon displacement as $\delta l = J^{\#} \cdot (t_{ref} - t)$. The damping parameter is set to $\lambda = 0.1$ and γ to 3 mm.

The data-driven controller computes the inverse mapping as:

$$\hat{J}_{t2l} = \frac{\partial N_l^1}{\partial t}\Big|_t \tag{7}$$

and avoids the use of encoders, enabling a fully self-contained vision-based pipeline at the cost of propagating camera estimation errors through the Jacobian. Conversely, combining both sensing modalities in the model-based controller improves accuracy by compensating for their respective limitations.

In both cases, the trajectory from the current position t to the target t_{goal} is generated by a straight-line Cartesian path planner $\mathcal{W}^1$ with waypoint spacing $\epsilon = 1$ mm. The sequence is dynamically updated based on camera feedback. The same algorithms are also employed in open-loop control as a baseline, where robot states are internally updated through either model-based or data-driven forward kinematics.

3.3.2 | Multi-Section Controller

The multi-section continuum arm is inherently redundant. The proposed model-based null-space controller exploits the Jacobian to optimize secondary objectives within the null space associated with the pose task. The general resolved motion rate solution [19] is:

$$\dot{l} = J^{\#}\dot{p} + N\xi_l \tag{8}$$

where $J^{\#} = J^T(JJ^T + \lambda^2 I)^{-1}$ is the regularized Moore-Penrose pseudoinverse, and $N = I - J^{\#}J$ projects ξ_l onto the null space of J. We minimize the quadratic cost function:

$$H(l) = \frac{1}{2}(l - \bar{l})^{\mathrm{T}}(l - \bar{l}) \tag{9}$$

where $\bar{l}$ denotes the mean tendon actuation estimated from the dataset. Reducing the deviation from mid-range values keeps the manipulator away from its physical limits, preserving flexibility for future tasks, and promotes smooth, coordinated joint motions

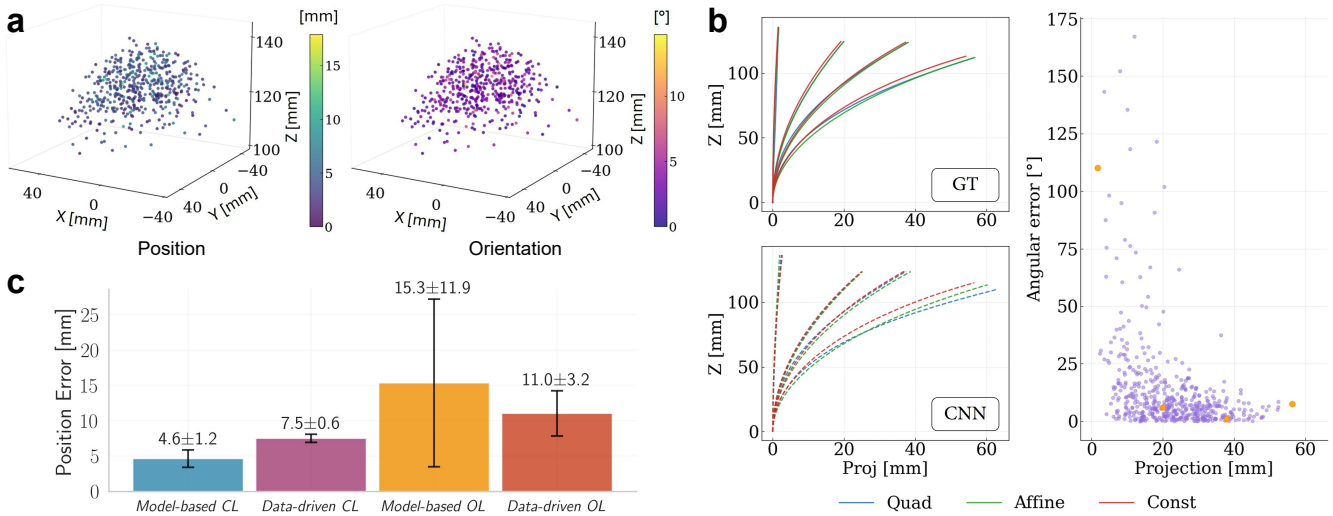

FIGURE 5 | (a) Distribution of position (mm) and orientation (°) reconstruction errors for the single-section arm on the test set. (b) Single-section curvature reconstruction and bending angle error. Four test examples with increasing deformation are shown, with GT (ground-truth) reconstructions as solid and CNN predictions as dashed lines. Axes scaling is not uniform to emphasize the bending effect. The corresponding ϕ angular errors are highlighted in orange. (c) Performance of the proposed controllers on the control workspace of the single-segment arm, reporting both average errors and standard deviations.

by distributing control effort evenly across the tendons. The secondary objective is enforced by adding the projected gradient of $H(l)$ [35] to the control law:

$$\dot{l} = J^{\#}\dot{p} - N \cdot \nabla H = J^{\#}\dot{p} + N(\bar{l} - l) \tag{10}$$

Algorithm 1 Model-based feedback controller

Set target p_{goal} and convergence threshold γ
Measure current pose p from $(f_p^3(I))$
Measure tip positions $[t_I, t_{II}, t_{III}]$ from $(f_p^3(I))$
Estimate $l \leftarrow l(t_I, t_{II}, t_{III})$ via PCC
while $d_{SE(3)}(p, p_{goal}) > \gamma$ do
  Compute next reference $p_{ref} \leftarrow \left(\mathcal{W}^3(p, p_{goal})\right)_1$
  Measure l from motor encoders
  Compute Jacobian $J \leftarrow J_{l2p}(l)$
  Compute damped pseudoinverse $J^{\#} \leftarrow J^{\mathrm{T}}\left(JJ^{\mathrm{T}} + \lambda^2 I\right)^{-1}$
  Compute null-space projector $N \leftarrow I - J^{\#}J$
  Move motors by $\delta l \leftarrow J^{\#} \cdot (p_{ref} - p) + N \cdot (\bar{l} - l)$
  Measure p from $(f_p^3(I))$
end while

Trajectories between the current pose p and the target p_{goal} are generated by an $SE(3)$ planner $\mathcal{W}^3$, interpolating positions linearly and orientations via SLERP on the $SO(3)$ manifold. Distance metrics [36] and parameters are detailed in the Supporting Information. The model-based controller is summarized in Alg. 1. Parameters λ and ϵ match the single-section values; the convergence threshold is set to $\gamma = 10$ mm, and the PCC-based Jacobian is estimated from encoders.

Similarly, the proposed data-driven controller reduces the deviation of joint movements from their mid-range values. Redundancy optimization is incorporated during training through a Lagrange multiplier $\zeta = 0.01$, which balances primary and secondary tasks. The inverse kinematics function $N_l^3(p)$ is learned via a two-stage scheme. First, a forward network $N_p^3(l)$ is trained to map actuator commands to end-effector poses. This network is then frozen and used to supervise the training of the inverse mapping $N_l^3(p)$. The loss combines a reconstruction error and a secondary objective:

$$\mathcal{L} = \|p - N_p^3(N_l^3(p))\|_2^2 + \zeta \cdot \|\bar{l} - N_l^3(p)\|_2^2 \tag{11}$$

The first term enforces task accuracy, while the second biases the solution toward $\bar{l}$. The secondary objective can be flexibly modified for alternative tasks such as obstacle avoidance or grasping. The data-driven controller is summarized in Alg. 2. As in the single-section case, the Jacobian is estimated directly from the pose without encoder feedback.

Algorithm 2 Data-driven feedback controller

Set target p_{goal} and convergence threshold γ
Measure current pose p from $(f_p^3(I))$
while $d_{SE(3)}(p, p_{goal}) > \gamma$ do
  Compute next reference $p_{ref} \leftarrow \left(\mathcal{W}^3(p, p_{goal})\right)_1$
  Compute $\hat{J}_{p2l} \leftarrow \left.\frac{\partial N_l^3}{\partial p}\right|_p$
  Move motors by $\delta l \leftarrow \hat{J}_{p2l} \cdot (p_{ref} - p)$
  Measure p from $(f_p^3(I))$
end while

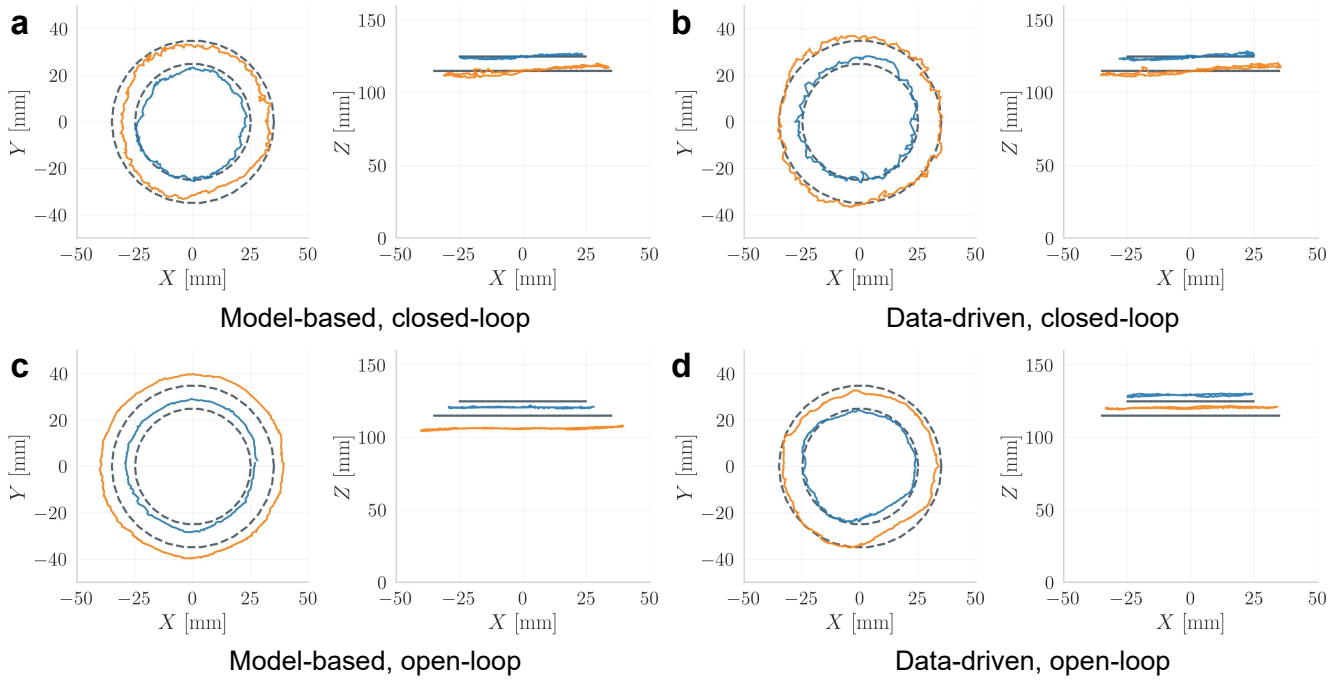

FIGURE 6 | Trajectory tracking for the single-section soft manipulator. The dashed grey line indicates the reference trajectory; executed trajectories are shown in orange for the lower circle and blue for the upper one. (a) Model-based feedback controller. (b) Data-driven feedback controller. (c) Model-based feedforward controller. (d) Data-driven feedforward controller.

4 | Experimental Results

We evaluate the proposed framework on both single- and multi-section soft manipulators. First, we evaluate tip pose estimation and curvature reconstruction from embedded cameras with CNNs, considering quadratic, affine, and constant curvature models. We further assess the data-driven kinematics in comparison to the PCC approximation. Then, we benchmark the proposed model-based and data-driven controllers on target reaching and trajectory tracking, and demonstrate the advantages of closed-loop control under external disturbances.

4.1 | Single-Segment Reconstruction and Control

4.1.1 | Static Pose Reconstruction

We embed a single camera at the base to estimate both tip position and orientation. The final position accuracy corresponds to 2.66± 1.49% of the robot length (a translation error of 3.99 ± 2.23 mm), and the overall rotation error is 3.11° ± 1.96°. The reconstruction error distribution is reported in Fig. 5**a**. Overall, accuracy improves in bent configurations, where helicoidal deformations provide stronger visual cues for state estimation. Compared to IMU-based approaches [6], which have reported degradation at high bending, these results suggest that embedded vision may be advantageous at workspace boundaries.

4.1.2 | Curvature Reconstruction

The quadratic, affine, and constant curvature CNN models were trained and tested independently, each predicting $\hat{\phi}$, $\hat{L}$, and the corresponding polynomial curvature parameters. Fig. 5**b** shows four representative configurations with increasing deformation projected onto the bending plane. Ground-truth reconstructions are compared with predictions from the three curvature models. The corresponding ϕ angular errors are plotted against the backbone projection on the base plane. The discrepancy among the three ground-truth curvature models grows with deformation due to actuation non-idealities and self-weight. CNN predictions remain accurate even in boundary configurations. However, angular error in ϕ can reach nearly 180° near straight configurations, where such precision is not needed for accurate reconstruction, as backbone error remains limited. As bending increases, ϕ error decreases significantly.

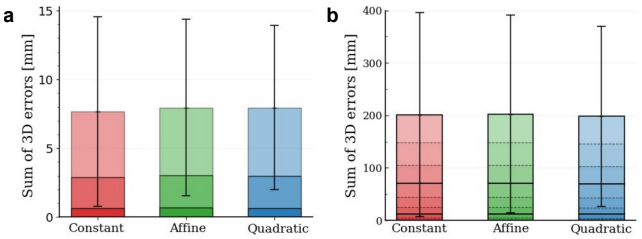

FIGURE 7 | Cumulative 3D reconstruction error of the curvature models. (a) Single-section arm: reference points at $\frac{L}{3}$, $\frac{2L}{3}$, and L. (b) Multi-section arm: reference points placed every ninth of the manipulator length. Error bars indicate ± one standard deviation. Note the different vertical scales.

To explore how the internal camera can be used for distributed proprioception, we evaluate reconstruction accuracy at three reference points on the backbone, respectively at $\frac{L}{3}$, $\frac{2L}{3}$, and L. Fig. 7**a** reports the average 3D error for each point, as well as the cumulative standard deviations across all curvature models. Although

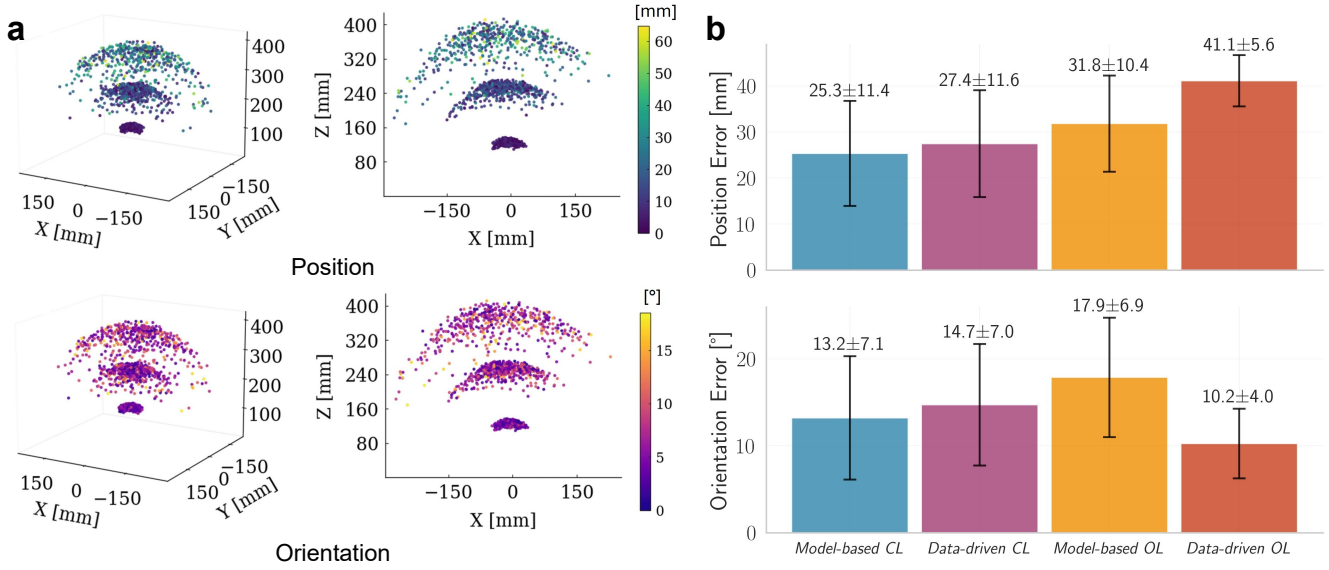

FIGURE 8 | (a) Distribution of position (mm) and orientation (°) reconstruction errors for the multi-section arm on the test set. (b) Performance of the four proposed controllers on the control workspace of the multi-segment arm, reporting average errors and standard deviations for translation and rotation accuracy.

mean values are comparable, the standard deviation decreases with increasing model descriptiveness.

4.1.3 | Data-Driven Kinematics

The inverse kinematics network N_l^1 provides a 3-to-3 mapping between tip position and actuation space. It achieves 0.86 ± 1.12 mm tendon error, outperforming the PCC approximation (1.99 ± 1.85 mm).

4.1.4 | Control Performance on Target Points

Twenty workspace targets distributed via k-medoids clustering were reached sequentially by all controllers; each target was visited once, and errors are reported as the final position error at convergence. Results are shown in Fig. 5**c**. Closed-loop controllers approach the perception baseline: model-based and data-driven controllers yield 3.1 ± 0.8% and 5.0 ± 0.4% position error, respectively. The complementary behavior of vision (accurate at high bending) and encoder-based PCC (accurate near straight) leads to the best combined performance. The learning-based algorithm is more consistent but likely accumulates perception noise when evaluating the Jacobian directly from the estimated tip position. Conversely, the data-driven feedforward controller (7.3 ± 2.1%) outperforms its model-based counterpart (10.2 ± 7.9%) due to more accurate kinematics. Details on the k-medoids clustering algorithm, together with each closed-loop controller's error distribution, are provided in the Supporting Information.

4.1.5 | Control Performance on Trajectories

Two circular trajectories ($z = 115, 125$ mm; radii 35, 25 mm; 30 waypoints each), at different heights and dimensions to test bending and compression, were tracked by all controllers; results are shown in Fig. 6. Closed-loop controllers successfully reproduce both circles; the model-based approach drifts inward at high compression, while the data-driven controller achieves higher overall accuracy. Open-loop performance is lower: the model-based controller overestimates the radius, and the data-driven one is more irregular but reaches the target points more closely.

4.2 | Multi-Segment Reconstruction and Control

4.2.1 | Static Pose Reconstruction

We embed three cameras at the base and at the interfaces between segments to estimate each section tip position and orientation. The pose errors estimated by the f_p^3 CNN are reported in Table 1. Position error for Module I (2.96 ± 1.79% of section length) is consistent with the single-segment case. It accumulates along sections for Modules II and III, reaching 4.67 ± 2.93% and 5.88 ± 3.66% up to the respective tips, while orientation remains below 9°.

TABLE 1 | Pose errors for the multi-section arm.

Section	Position [mm]	Orientation [°]
I	4.44 ± 2.68	6.78 ± 15.94
II	14.01 ± 8.79	6.45 ± 3.74
III	26.45 ± 16.49	8.74 ± 13.05

The overall distribution of pose reconstruction errors on the test set across the three section tips is illustrated in Fig. 8**a**. The point clouds for all tips are overlaid in both perspective and frontal views. Errors near the base are smaller than at the distal end, with accuracy decreasing along sections due to chained frame transformations. Bent configurations, however, partially offset accumulation by providing more distinctive image features.

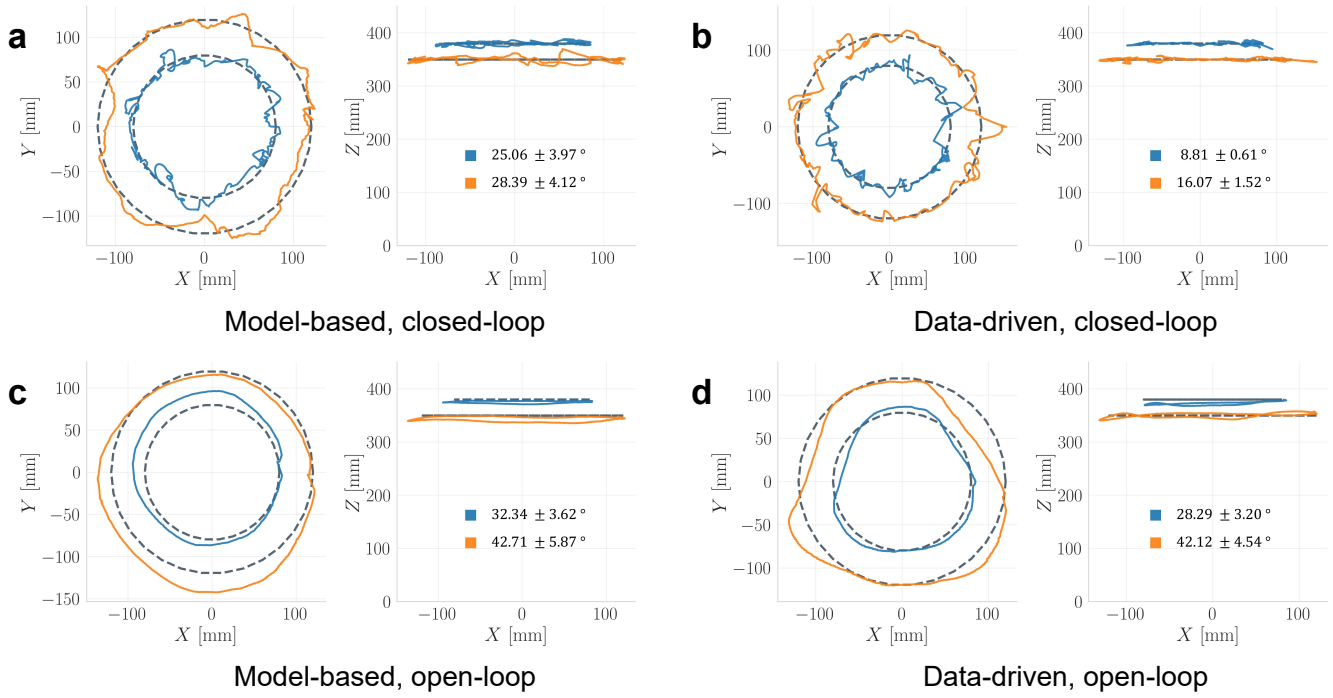

FIGURE 9 | Trajectory tracking for the multi-section soft manipulator. Angular values along both trajectories represent the rotation error on the corresponding circumferences, with smaller values on the inner circle and larger values on the outer one. (a) Model-based feedback controller. (b) Data-driven feedback controller. (c) Model-based feedforward controller. (d) Data-driven feedforward controller.

4.2.2 | Curvature Reconstruction

Three CNN models predict bending angles $\hat{\phi}_1, \hat{\phi}_2, \hat{\phi}_3$, central axis lengths $\hat{L}_1, \hat{L}_2, \hat{L}_3$, and the polynomial curvature parameters for each section. We evaluate reconstruction accuracy at nine uniformly spaced backbone reference points in Fig. 7**b**. Errors grow toward the distal tip; all models provide similar approximations, though increasing model complexity reduces variance.

4.2.3 | Data-Driven Kinematics

Forward kinematics accuracy is compared between the two approaches: the neural network (5.1 ± 3.3% position, 7.25 ± 3.88° rotation) outperforms PCC (8.9±6.0%, 21.47±23.05°) in both pose metrics. The composition $N_p^3(N_l^3(p))$ yields 2.1±1.2% translation and 8.3° ± 5.4° rotation error, though actual inversion accuracy combines this with the forward prediction error.

4.2.4 | Control Performance on Target Points

The same k-medoids algorithm used for the single-section arm defined 20 target poses. Control performance is shown in Fig. 8**b**. Closed-loop controllers achieve near-sensing accuracy; the model-based controller reaches 5.6 ± 2.5% position error and the data-driven one 6.1±2.6%, with differences linked to whether the Jacobian uses encoder or camera feedback. Open-loop performance degrades: PCC yields 7.1 ± 2.3% while the data-driven approach reaches 9.1 ± 1.2% due to compounding kinematic errors, though with superior orientation tracking and lower variance than the model-based open-loop controller.

4.2.5 | Control Performance on Trajectories

Two circular trajectories with horizontal end-effector orientation ($z = 350, 380$ mm; radii 120, 80 mm; 30 poses each) were tracked. The end-effector was kept horizontal as this pose is underrepresented in the training set and more challenging to generalize. Fig. 9 shows the results. Closed-loop controllers successfully reproduce both trajectories; the model-based approach tracks better in the circle plane, while the data-driven controller provides superior orientation and height stability. Open-loop controllers show larger errors and lower rotation accuracy. Compared to the single-section case, feedforward accuracy is lower, as maintaining horizontal orientation requires precise multi-section coordination that is unreliable without feedback.

4.2.6 | External Disturbances

TABLE 2 | Control performance comparison with external payload.

Control	Translation error [mm]	Rotation error [°]
Closed-loop	28.55 ± 14.40	14.39 ± 6.28
Open-loop	64.56 ± 15.96	25.16 ± 9.25

A 50 g payload (∼ 40% of effective robot mass, including TPU body, fabric, cameras, and PLA interfaces) was attached to the tip, emulating a sustained tip contact force and serving as a proxy for evaluating closed-loop robustness under persistent external interactions. Given its superior performance in the unloaded

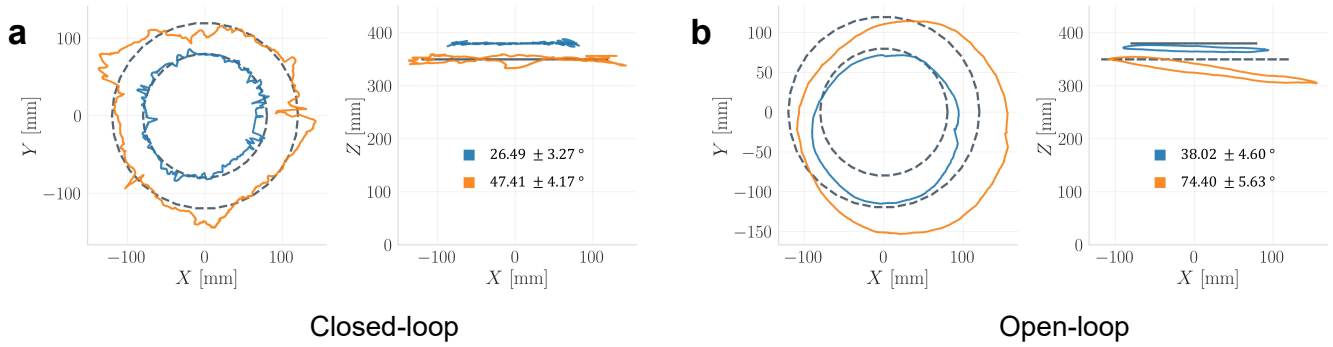

FIGURE 10 | Trajectory tracking for the multi-section robot under an external payload of 50 g attached at the tip, corresponding to about 40% of the effective mass of the manipulator. (a) Model-based feedback controller. (b) Model-based feedforward controller.

experiments, the model-based controller was selected for this evaluation. Table 2 compares 20-target tracking error under load (error distribution in the Supporting Information). The closed-loop controller maintains 6.3±3.2% translation error and rotation below 15°, compared to 14.5 ± 3.5% and larger orientation deviations for open-loop. Fig. 10 shows trajectory tracking under disturbance: the closed-loop controller recovers from payload-induced deviations even with a residual rotational bias, while open-loop control fails to maintain the desired path in both radius and height, becoming unbalanced toward the side of the off-center weight attachment.

5 | Conclusion and Discussion

This work proposes a scalable framework for vision-based proprioception and control of soft continuum robotic arms. Miniature cameras embedded in each segment feed deep learning algorithms that reconstruct the full shape of the robot, enabling closed-loop control strategies, both model-based and data-driven, which explicitly exploit kinematic redundancy to accomplish secondary tasks within the null space.

Utilizing two tendon-driven continuum manipulators, we propose a minimally invasive and compliance-preserving design using a flexible opaque covering that scales along the arm length without affecting its behavior. Experimental results demonstrate that our sensing approach reaches competitive performance in reconstructing both tip pose and full-body curvature, with preserved accuracy at workspace boundaries. Closed-loop control achieves near-sensing performance and outperforms open-loop alternatives, while being robust to external disturbances.

This sensing and control framework endows soft robots with accurate motion capabilities. Leveraging their redundancy, continuum manipulators can accomplish complex tasks across multiple domains. For instance, they can be employed in search-and-rescue or inspection within confined and cluttered environments, or in minimally invasive surgery. In such settings, full-body curvature feedback from embedded cameras enables null-space control to actively shape the arm for obstacle avoidance while maintaining the desired tip pose.

This strategy has some limitations, which may be addressed in future work. At the sensing level, the embedded vision system currently relies on ambient lighting provided by the external environment. Illumination passes through the opaque covering, and variability is mitigated through image augmentations during training. To enable operation in poorly lit or nocturnal environments, the system should be extended with miniature internal light sources. This paper primarily investigates vision alone, integrating encoder signals for Jacobian estimation. Unlike external cameras, the internal setup avoids occlusion and fixed-placement constraints, at the cost of lower update frequency. The 8 Hz update rate limits the framework to quasi-static operation, suitable for applications such as inspection or surgical guidance; higher rates could be achieved by fusing camera estimates with encoder feedback through temporal filtering. Future work should integrate the camera-based solution with alternative sensing strategies to improve configuration accuracy. Moreover, the system does not detect contact or force information, although cameras could provide both. Extending proprioception to capture surface interactions through training or simulation would enhance robot capability, while force detection would add a feedback channel for closed-loop control in dynamic conditions. On the control side, future work will extend null-space control to additional secondary objectives. In this paper, redundancy was exploited to balance tendon effort, but the same methodology applies to shaping arm configurations for obstacle avoidance while tracking a desired end-effector trajectory. Furthermore, null-space control could be leveraged to wrap around objects and enable grasping, using curvature feedback to regulate how the robot conforms to complex geometries.

Author Contributions

Conceptualization: L.V., G.P., and J.H. **Methodology**: L.V. **Investigation**: L.V. **Writing**: L.V. and J.H. **Review and editing**: L.V., F.B., and J.H. **Supervision**: G.P., F.B., and J.H.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The code supporting this study is openly available at https://github.com/CREATE-Lab-EPFL/VisionSoftControl.git.

The dataset is available on Zenodo at https://doi.org/10.5281/zenodo.20138549.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.